Constrained minimization of the ghost-GA functional at fixed $|R|$

(Hubbard dimer, direct minimization without the impurity problem)

November 6, 2026

This note describes how the gGA solution of the half-filled Hubbard dimer is recomputed with the renormalization vector R_i of each fragment held at a prescribed length. It is the method behind Sec. "Ionic weight of Ψ at fixed $|R|$" of `results.tex`. No impurity problem is solved and there is no self-consistency loop: the energy functional of the direct minimization is minimized under its own constraints plus one extra equality constraint on $|R_i|$.

1 The functional that is minimized

Fragment $i = 0, 1$ carries a normalized many-body state ϕ_i of the embedding space, with $N = 1+N_g$ quasiparticle/bath orbitals per spin. The variables are the unnormalized coefficient vectors x_0, x_1 of the two fragments ($\phi_i = x_i/|x_i|$) and a real $2N \times N$ matrix X whose column space is the occupied space of the Slater determinant ψ^*; its one-body density matrix per spin is $\rho = X(X^TX)^{-1}X^T$. All of these are stacked in one vector z (code: `model.unpack(z)` $\to (x_0, x_1, X)$, class `NotImpDimer` of `not_imp_gga_dimer.py`).

From ϕ_i one builds the density matrix $\Delta_i^{bb}(\phi_i)$ and the vector $D_i(\phi_i)$ (spin averages of $\langle\phi_i|d_a^\dagger d_b|\phi_i\rangle$ and $\langle\phi_i|d_a^\dagger c|\phi_i\rangle$), and the renormalization vector

$$R_i(\phi_i) = \left[\Delta_i^{bb}(1 - \Delta_i^{bb})\right]^{-1/2} D_i, \tag{1}$$

which in the code is `frag(x,i)["R"]` (a small shift, 10^{-9}, regularizes the inverse square root). It has N components. The energy is

$$E(z) = -4t\, R_0^T \rho_{01} R_1 + \sum_i \langle\phi_i|H_{\rm loc}^{(i)}|\phi_i\rangle, \tag{2}$$

with ρ_{01} the block of ρ connecting the two fragments and $H_{\rm loc}^{(i)} = Un_\uparrow n_\downarrow - \mu_i n$, $\mu_i = U/2 \pm \Delta V$. The standard gGA equality constraints are, for each fragment and each $a \le b$,

$$c_{i,ab}^{\rm gGA}(z) = \rho_{ii,ab} + \Delta_{i,ab}^{bb}(\phi_i) - \delta_{ab} = 0, \tag{3}$$

(code: `model.constraints`, with analytic Jacobian `model.constraint_jac`). The unconstrained gGA solution minimizes Eq. (2) under Eq. (3) with SLSQP; call it $z^{\rm sol}$ and its renormalization vectors $R_i^{\rm sol} = R_i(z^{\rm sol})$.

2 What is fixed: $|R_i|^2$, not the vector

For $N_g = 0$, R_i is a single number ($R_i = \sqrt{Z_i}$), so fixing R_i^2 fixes R_i itself.

For $N_g \geq 1$ the functional is invariant under orthogonal rotations of the quasiparticle basis of each fragment: such a rotation turns R_i into another vector of the same length and leaves E and the constraints unchanged. Therefore only $|R_i|$ is a variable with a meaning. Fixing the whole vector R_i at a value of unchanged length returns the same energy and the same Ψ (checked at $U/t = 4$, $N_g = 2$: $E_{\rm var} = -4.511136$ and $P_{\rm ion} = 0.1453$ for all s; this is the option `rot` of the script). The constraint used is therefore, for $i = 0, 1$,

$$g_i(z) \equiv R_i(\phi_i)^T R_i(\phi_i) - s^2 \left|R_i^{\rm sol}\right|^2 = 0, \tag{4}$$

where s is the scale factor ($s = 1$ is the unmodified solution). Two scalar equations are added to the gGA constraints of Eq. (3). In the code, `Rfun(z)` returns $(R_0^T R_0,\, R_1^T R_1)$, the target is `Rt` $= s^2(|R_0^{\rm sol}|^2, |R_1^{\rm sol}|^2)$, and the constraint is `Rfun(z) - Rt`.

3 The constrained problem and its solver

For a given s the problem is

$$\min_z \; E(z) \qquad \text{subject to} \qquad c^{\rm gGA}(z) = 0, \quad g(z) = 0, \tag{5}$$

solved with `scipy.optimize.minimize(..., method="SLSQP")`, with the analytic gradient of E (`model.energy(z)` returns E and $\partial E/\partial z$), the analytic Jacobian of $c^{\rm gGA}$, and for g a central finite-difference Jacobian (step $h = 10^{-6}$ per component of z; `Rjac`). The options are `ftol`$= 10^{-13}$ and `maxiter`$= 1000$. The solution of the constrained problem is a point that satisfies the gGA constraints and has the prescribed $|R_i|$, and it minimizes the same functional as the solution; it is not a solution of the gGA stationarity equations (the energy is higher than at $s = 1$). Since the functional is the one of the direct minimization, no impurity Hamiltonian and no outer loop over λ or V appear.

4 Continuation in s

The unconstrained solution is the starting point and s is changed step by step, each point started from the previous converged one:

1. $s = 1$: the stored solution $z^{\rm sol}$ is taken as is (it is saved as the first point). As a check, $R_i(z^{\rm sol})$ is compared with the R_i derived from the stored solution files.
2. From $s_{\rm cur}$ the next value is $s_{\rm new} = s_{\rm cur} \pm h$, with $h \leq ds$ ($ds = 0.05$ for $N_g = 0$ and 0.02 for $N_g = 2$ in the runs of `results.tex`), walking once upward (to $s_{\rm max}$) and once downward (to $s_{\rm min}$), both starting from $s = 1$.
3. The constrained problem is solved starting from the last accepted z. The point is accepted if SLSQP reports success and the largest violation of all constraints, $\max\left(|c^{\rm gGA}|, |g|\right)$, is below 10^{-7}. Then z is saved at once (`Rfixed/ng<Ng>_<tag>_U<U>_norm/s_<s>/z.npy`), the reference point is moved to $s_{\rm new}$ and the step is increased, $h \to \min(1.5h, ds)$.
4. Otherwise the step is halved, $h \to h/2$, and the attempt is repeated from the last accepted point. If $h < 10^{-3}$ the walk in that direction stops and the last accepted s is the limit reached.

In this way values of $s > 1$ were reached for $N_g = 0$ (up to 1.1 for $U/t \geq 4$), while for $N_g = 2$ no value $s > 1$ converged (at $U/t = 4$, $s = 1.1$ does not converge), so $s \leq 1$ is studied.

5 From the constrained point to Ψ and $P_{\rm ion}$

Every accepted point is passed to the construction of the Gutzwiller wave function Ψ (`gutzwiller_psi.build_psi`, Sec. "The Gutzwiller wave function itself" of `results.tex`), and `analyse` gives $P_{\rm ion}$, the energy E_Ψ and the overlap with the FCI ground state. One line per point is appended to `psi_scan/Rfix_ng<Ng>_<tag>_U<U>_n`

$$s,\ E_{\rm var},\ \text{constraint},\ |R_0'|,\ P_{\rm ion},\ E_\Psi,\ \text{overlap},\ \left(|R_\Phi|^2 - |R_0'|^2\right),$$

where $E_{\rm var} = E(z)$ is the value of Eq. (2) at the constrained point and the last column is the difference between $|R|^2$ computed from the Φ-matrix in the construction of Ψ and the target (of order 10^{-7} or smaller in the $N_g = 2$, $U/t = 4$ data).

Script. `scripts/psi_fixedR.py <dV0|dV2|dV2n> <Ng> <U> <none|rot|norm> [s_min s_max ds]`; the results of `results.tex` use the option `norm` (Eq. (4)) at $\Delta V = 0$.